

8BE030 Registration Assignment

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Group 1

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Introduction

Image registration is a processing step that is used to analyse medical images such that two or more images can anatomically aligned for better comparison. In this study, T1-weighted MRI brain scans were used, with two versions each slice: the original ('fixed') image, and a transformed ('moving') image.

MRI scans are prone to Gaussian noise, resulting in reduced image contrast and hampered feature detection [1]. As noise is a realistic complication in medical imaging, the aim of this project was to investigate its effect on intra-modal affine registration. Two similarity metrics were considered: normalised cross correlation (NCC) and mutual information (MI). NCC measures the linear relationship between pixel intensities and is considered better suited to intra-modal registration due to this linearity assumption [2]. MI measures statistical dependence between intensity distributions [2] and while it can be applied to intra-modal registration, it is reported to perform more poorly in this context due to local maxima interfering with optimisation [3].

The research question was therefore: *What impact does Gaussian noise have on intra-modal image registration, and can a conclusion be made on whether affine NCC or MI performs better?*

It is hypothesised that Gaussian noise will lead to poorer registration in general. Although NCC would ordinarily be expected to perform better than MI at intra-modal registration, noise may interfere with the linear pixel-intensity relation that it relies upon. While MI does not depend on this linear relation, it is considered better for inter-modal registration. It is therefore expected that both metrics will perform similarly, as they both face a disadvantage under the conditions of this study.

Study design

Eight fixed and moving T1-weighted MRI scans were used in this study: 1_2_t1(_d) to 3_3_t1(_d). The first T1 scan was excluded due to erratic results (see Jupyter notebook for more information). Gaussian noise was added to the moving images, where the noise parameters were chosen to simulate moderate noise while still preserving image features, based on trial and error. The learning rate, number of iterations and initialisation vector were also set as independent variables, based on the initial investigations of Task 1. By keeping these parameters fixed, any differences in registration between NCC and MI were assumed to be due to the similarity metric itself. Thus, each image pair was evaluated using both similarity metrics under identical conditions.

Methods

Gaussian noise ($\mu = 0, \sigma = 20$) was added to the moving images before registration. Affine NCC and MI were performed for each image pair with fixed learning rate ($LR = 0.0001$), number of iterations ($N_{iter} = 200$) and initialisation vector ($\vec{x} = [0, 1, 1, 0, 0, 0, 0]$). Affine registration was chosen because it was not known what random transformations had been applied to the moving image. Affine was therefore more suitable as it would be able to handle both rigid and affine transformations.

The registration runs by updating the transformation vector with each iteration in order to improve the alignment between the fixed and noisy moving images. After registration, the alignment was assessed. Since MI and NCC similarities cannot be quantitatively compared, results were assessed in two ways. The first was visual inspection of the final overlaid images. Second, the mean-squared error (MSE) was found between the fixed image and the registered noisy moving image by comparing both their pixel arrays.

Results

For each image pair, the mean squared error between the final transformed image (the noisy moving image) and the fixed image can be seen in [Table 1](#). The numbers themselves are not particularly indicative, but the MSE for each similarity metric per image pair can be compared to assess how similar the registrations were. As such, the difference between the MI MSE and NCC MSE was also computed. A positive difference meant that NCC had a lower MSE than MI, suggesting a better alignment. However, based only on MSE and a limited sample size an objective conclusion on alignment quality cannot be drawn.

It can be seen that both registrations performed similarly in most cases. The notable outliers are 1_2_t1, 2_1_t1 and 3_3_t1, where NCC far out-performs MI based on the MSE metric.

Table 1: MSE values for affine NCC and MI registration across image pairs

Image pair	NCC MSE	MI MSE	Δ MSE (MI - NCC)
1_2_t1 & 1_2_t1_d	860	2464	+1604
1_3_t1 & 1_2_t1_d	2721	2728	+7
2_1_t1 & 2_1_t1_d	2347	2388	+41
2_2_t1 & 2_2_t1_d	2854	2853	-1
2_3_t1 & 2_3_t1_d	2301	2308	+7
3_1_t1 & 3_1_t1_d	2664	2673	+9
3_2_t1 & 3_2_t1_d	3055	3047	-8
3_3_t1 & 3_3_t1_d	2510	2890	+380

A visual inspection of a case with similar MSEs can be seen in [Figure 1](#) and [Figure 2](#). The overlaid images are not very well-aligned, but both transformation vectors are in similar orders of magnitudes.

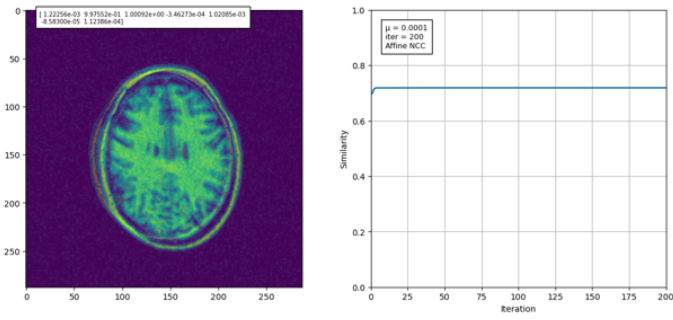


Figure 1: *Affine NCC, 2_3_t1, MSE = +7*

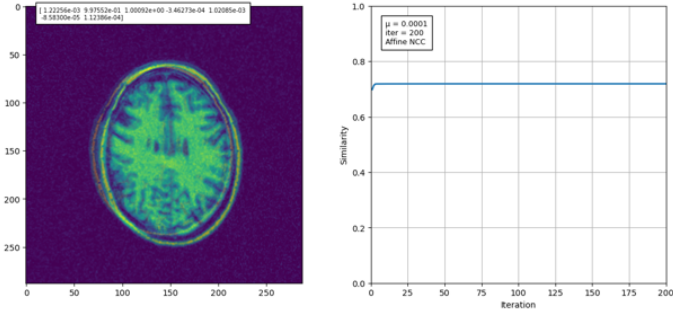


Figure 2: *Affine MI, 2_3_t1, MSE = +7*

A visualisation for the case with the most different MSEs can be seen in [Figure 3](#) and [Figure 4](#). NCC appears to have a much better alignment than MI here, explaining the much lower MSE.

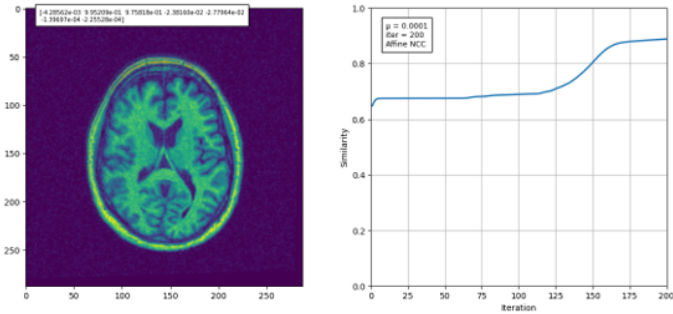


Figure 3: *Affine NCC, 1_2_t1, MSE = +1604*

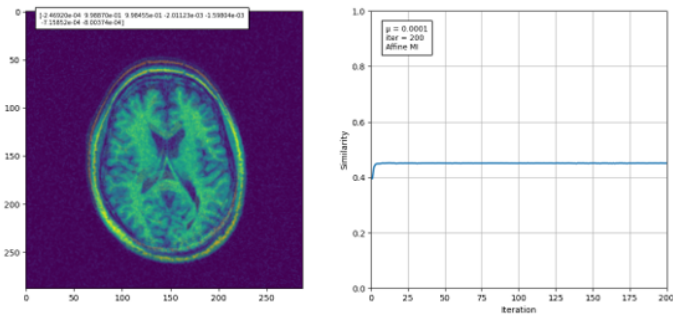


Figure 4: *Affine MI, 1_2_t1, MSE = +1604*

Discussion

The aim of this research was to compare affine intensity-based registration using NCC and MI after Gaussian noise was added to transformed T1-weighted images. The hypothesis was that Gaussian noise would degrade the registration performance of both metrics, and that MI and NCC would perform similarly. This was because the noise could affect the linear pixel-intensity relationship that NCC depends on, and because MI performs poorly on intramodal registration compared to intermodal. As such, both modalities were being tested outside their optimum conditions.

The results mostly supported this hypothesis. Across most image pairs, the MSEs for NCC and MI were close to each other. This suggested that neither similarity metric was clearly better than the other. Although NCC had a lower MSE in most image pairs, the differences were usually small.

However, there were a few notable exceptions where NCC far out-performed MI (based on the MSE metric). An example of this can be seen in [Figure 3](#) and [Figure 4](#). From a visual inspection, it can be seen that NCC has a much better alignment of the fixed and moving images compared to MI. It was also interesting to inspect the similarity plot, where the similarity score appeared to reach an initial convergence and then abruptly began increasing around 75 iterations. In contrast, the MI similarity stayed at the same value for nearly the entire iteration cycle.

A possible explanation for this could be that the addition of Gaussian noise makes the gradient ascent optimisation more prone to falling into a local maximum. This is supported by the observation that for nearly all image pairs, the overlaid images for both similarity metrics show the same misalignment pattern. This suggests that optimisation consistently converged on a sub-optimal solution. From visual inspection it can also be seen that despite the similarity plots converging, the anatomical alignment remains imperfect in these cases.

In the case of [Figure 3](#), it is possible that the learning rate and updated transformation vector came together favourably such that the optimiser escaped the local maximum and reached the global one. Furthermore, the similarity plot still had not plateaued by the 200th iteration, suggesting an even smaller MSE could still be obtained with more iterations. In this case, the Gaussian noise was not enough to affect the linear pixel-intensity relationship and NCC performed better than MI.

A pitfall of this study was that validation of the MSE differences was done qualitatively through visual inspection, rather than quantitatively. This could be improved by finding a way to normalise the MI similarity score such that both scores can be directly compared. Another limitation was that only one noise level was tested. Using multiple noise levels would make it possible to study whether NCC or MI becomes more reliable as the noise increases. Additionally, the only metric used for quantitative comparison (the MSE) was calculated using image intensity differences, which could also have been influenced by the added noise. A final limitation was that only one learning rate was used, which meant that it was not possible to test whether the misalignments in most of the registrations were indeed due to local maxima and if this would have improved with a smaller learning rate.

In summary, Gaussian noise reduced the performance of affine registration for both NCC and MI. Both metrics showed similar registration errors, so the hypothesis that their performance would be comparable under noisy conditions was somewhat supported. However, NCC performed better in a few notable cases, suggesting that the inter-play of learning rate with the transformation vector has a great impact on performance and requires further investigation. Therefore, a conclusion on whether NCC or MI performs better cannot be drawn without further study.

Contributions

<https://github.com/ariana-kdr/8be030-assignment1/tree/main>

Task 1 (in the notebook) was performed as a group.

Ariana Kadri: I was responsible for performing the affine intensity-based registration of a T1 and a T2 slice using MI as a similarity measure in Task 2 and writing an explanation. I also ran an extra rigid intensity-based registration of a T1 and a T2 slice using NCC, allowing for a more complete comparison of the different T1/T1, T1/T2, rigid/affine and MI/NCC combinations. For these registration tasks I made a modularised version of the given function `intensity_based_registration_demo()` that could take relevant inputs such that you did not need to make changes to the function itself. I then adapted the graph output so that more relevant information was displayed on it. This modularised function was then used by the rest of the group in their tasks, as well as in our open-ended project work. I then made a function for adding Gaussian noise to our moving image, which another group member built upon in another function. I ran 3 image sets using these functions and analysed the results, including the analysis for the outlier results. Additionally, I set up a Github repository for our group. Finally, I did a final edit on the report with a focus on tying the contents back to the research question.

Cecilia Yang: I was responsible for performing the affine intensity-based registration of two T1 slices using MI as a similarity measure in Task 2 and writing an explanation. Furthermore, I was responsible for writing code and creating functions that were used in the open assignment and I made the plan and task division for each group member. Additionally, I ran 4 images with this function and subsequently wrote up the interpretation and conclusion of these results. Finally, I cleaned up certain parts in the code and the notebook.

Chaymae Talhaoui: I was responsible for performing the affine intensity-based registration of a T1-weighted and T2-weighted image slice, using normalized cross-correlation as the similarity measure. This involved applying the registration function to align the two image slices and evaluating the resulting output. In addition, I contributed to the report by writing the introduction, study design, and methods sections. Furthermore, I ran the created registration function on one image and wrote a short interpretation of the output, describing whether the registration result was successful and how well the images were aligned.

Yousra Errajaji: I was responsible for performing the affine intensity-based registration of two T1 slices using NCC as similarity measure in Task 2 and writing an explanation. This required me to work with the registration function to align the two image slices. I also wrote the explanation of the results from the extra rigid intensity-based registration of a T1 and a T2 slice using NCC that Ariana performed and I wrote the overall conclusion of task 2. Additionally, I was tasked with going over all the answers in the notebook and rewriting/adjusting wherever necessary based on any new acquired information. For the report I was responsible for writing the results and discussion.

AI statement

ChatGPT was used as a writing support tool for the Introduction, Study Design, and Methods sections of this report. It was mainly used to improve wording, grammar, sentence structure, and clarity. It was also used to help reformulate some sentences in a more formal and scientific writing style.

Moreover, ChatGPT was used during the coding process to debug and clean up some parts of the code that were malfunctioning. (All AI generated suggestions were reviewed, edited, and approved before being included in the report).

Claude AI was used when modularising the given `intensity_based_registration_demo()` function to figure out the correct notation for passing the similarity functions (`reg.rigid_corr`, `reg.affine_corr`, `reg.affine_mi`) as input argument. Additionally, it was used to double check what python libraries the code needed at the end so the `requirements.txt` file could be updated correctly.

References

- [1] Goyal B, Dogra A, Agrawal S, Sohi BS. Noise Issues Prevailing in Various Types of Medical Images. Biomedical and Pharmacology Journal. 2018. Available from: <https://api.semanticscholar.org/CorpusID:54701555>.
- [2] Ghannam S, Abbott AL. Cross Correlation versus Mutual Information for Image Mosaicing. International Journal of Advanced Computer Science and Applications. 2013;4(11). Available from: <http://dx.doi.org/10.14569/IJACSA.2013.041113>.
- [3] Gao Z, Gu B, Lin J. Monomodal image registration using mutual information based methods. Image and Vision Computing. 2008;26(2):164-73. Available from: <https://www.sciencedirect.com/science/article/pii/S0262885606002642>.

Appendix

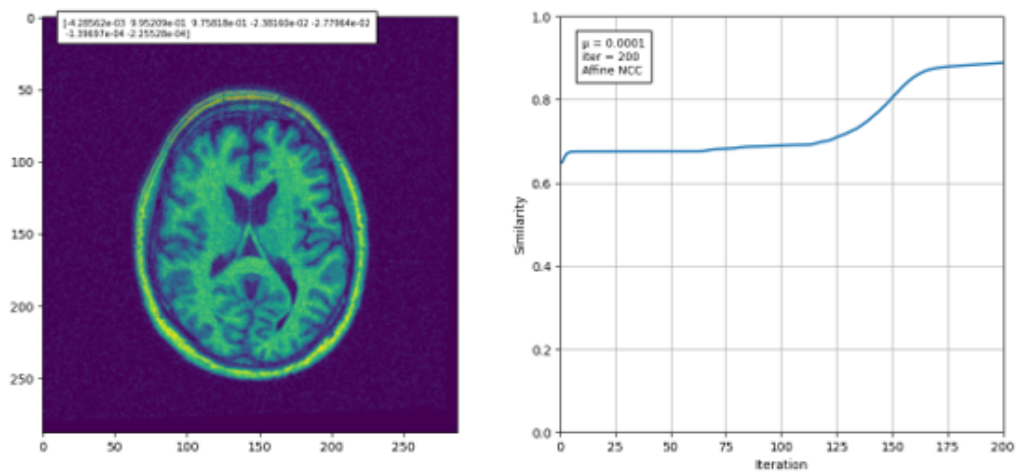


Figure 5: Affine NCC registration after Gaussian noise for 1_2_t1

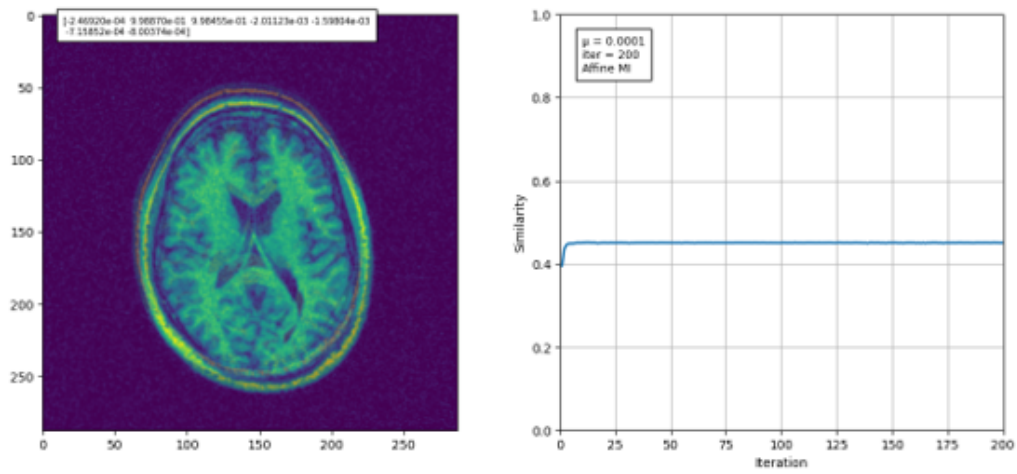


Figure 6: Affine MI registration after Gaussian noise for 1_2_t1

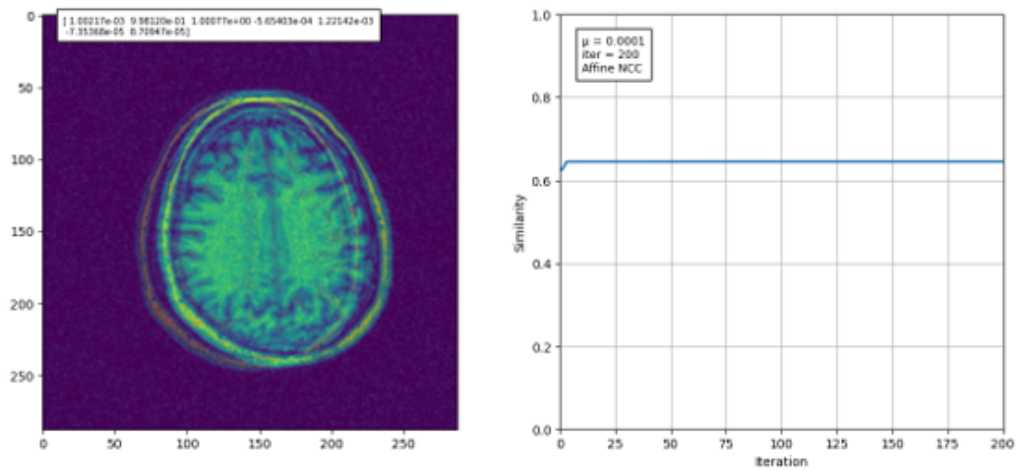


Figure 7: Affine NCC registration after Gaussian noise for 1_3_t1

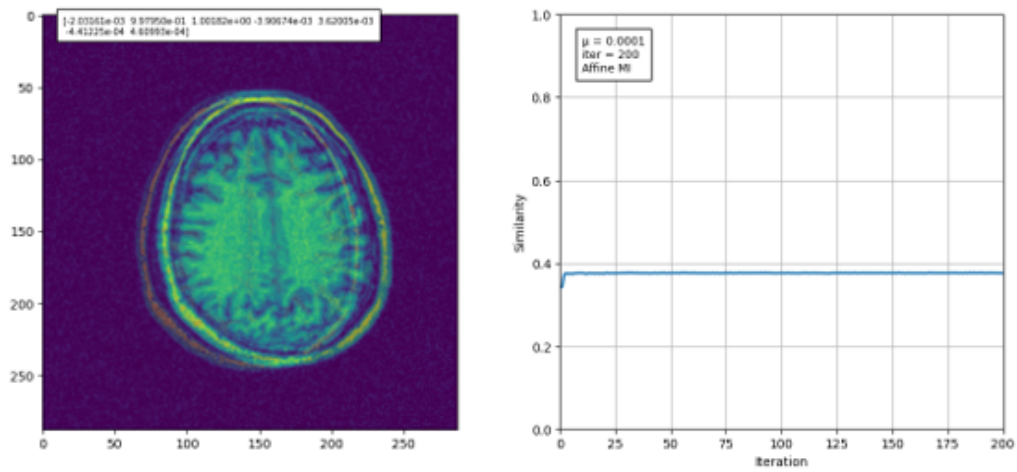


Figure 8: Affine MI registration after Gaussian noise for 1_3_t1

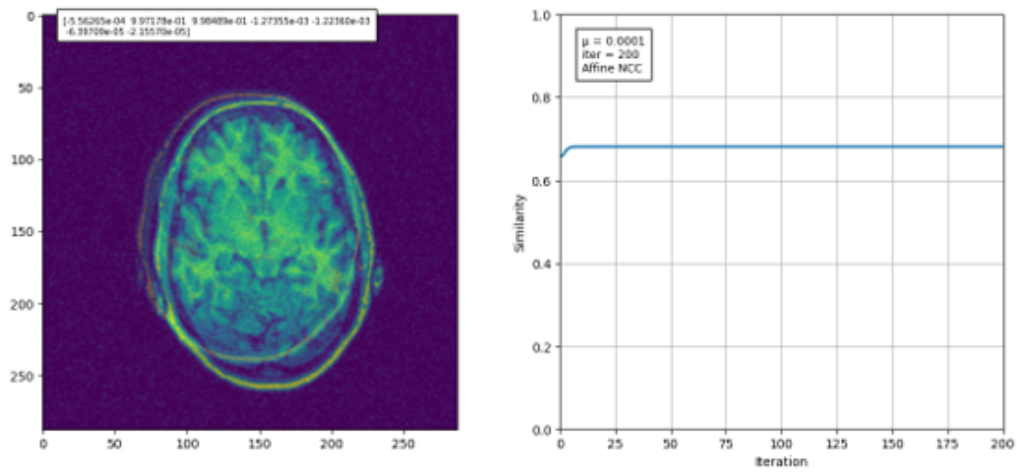


Figure 9: Affine NCC registration after Gaussian noise for 2_1_t1

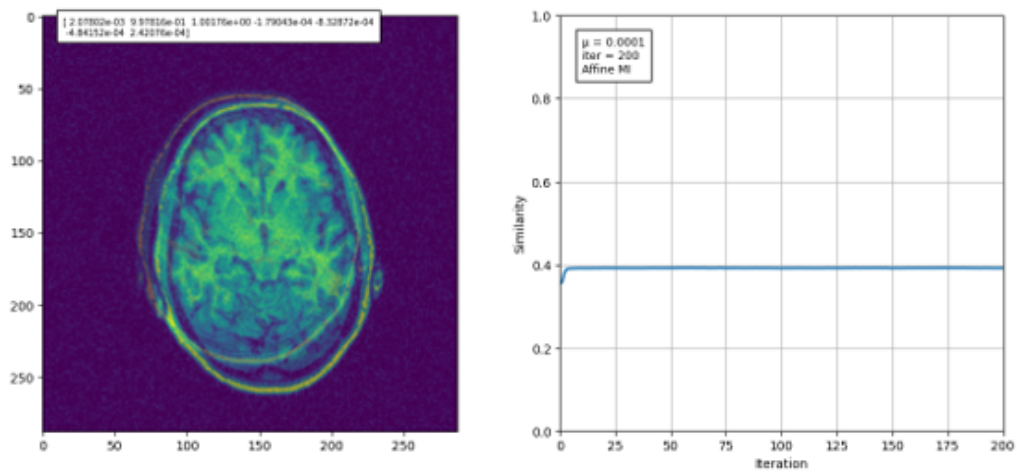


Figure 10: Affine MI registration after Gaussian noise for 2_1_t1

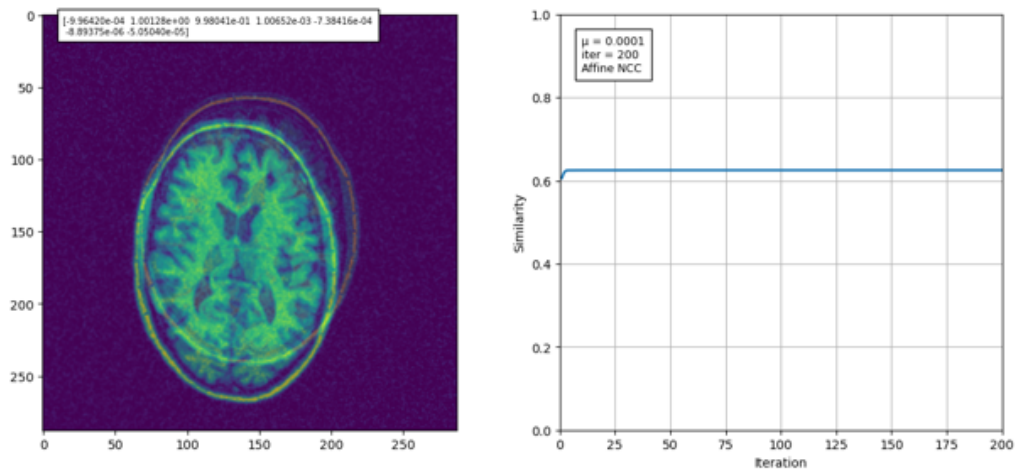


Figure 11: Affine NCC registration after Gaussian noise for 2_2_t1

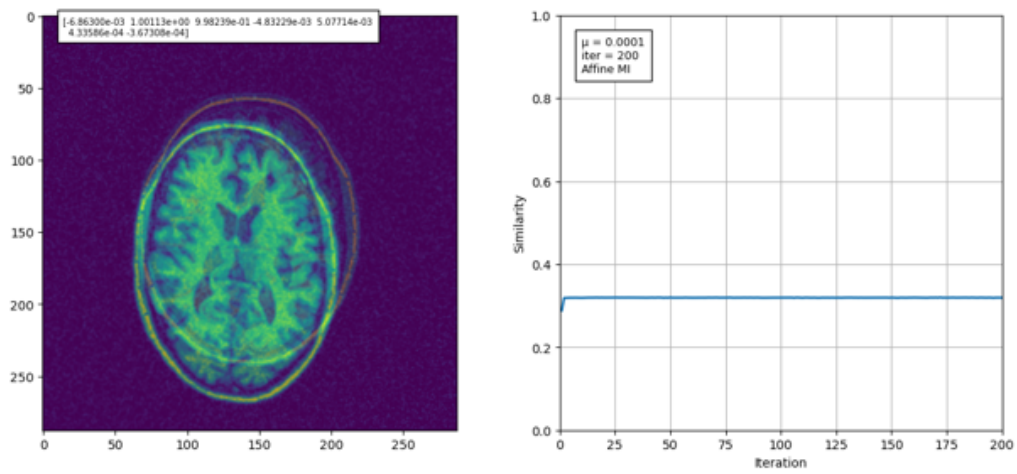


Figure 12: Affine MI registration after Gaussian noise for 2_2_t1

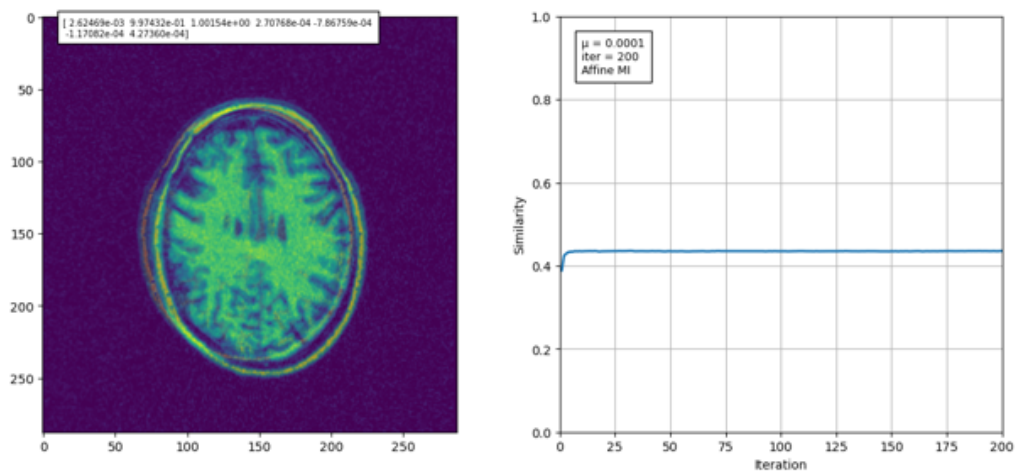


Figure 13: Affine NCC registration after Gaussian noise for 2_3_t1

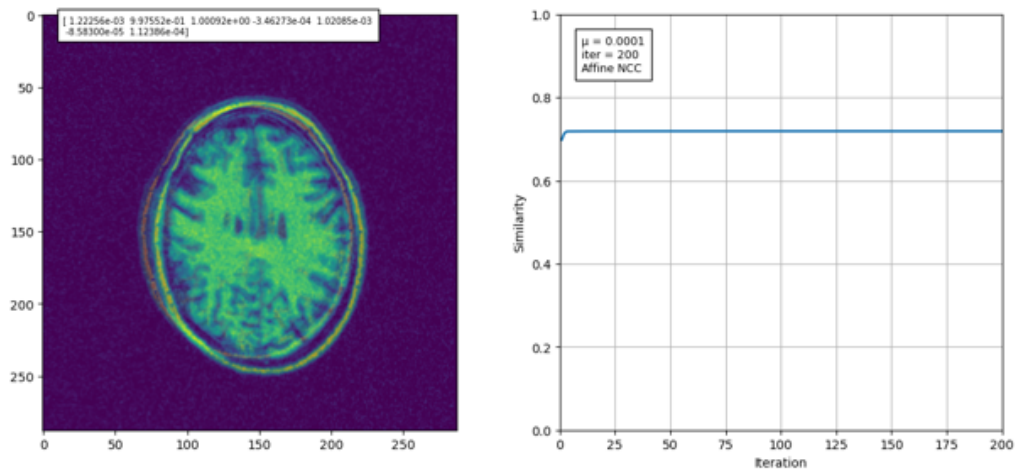


Figure 14: Affine MI registration after Gaussian noise for 2_3_t1

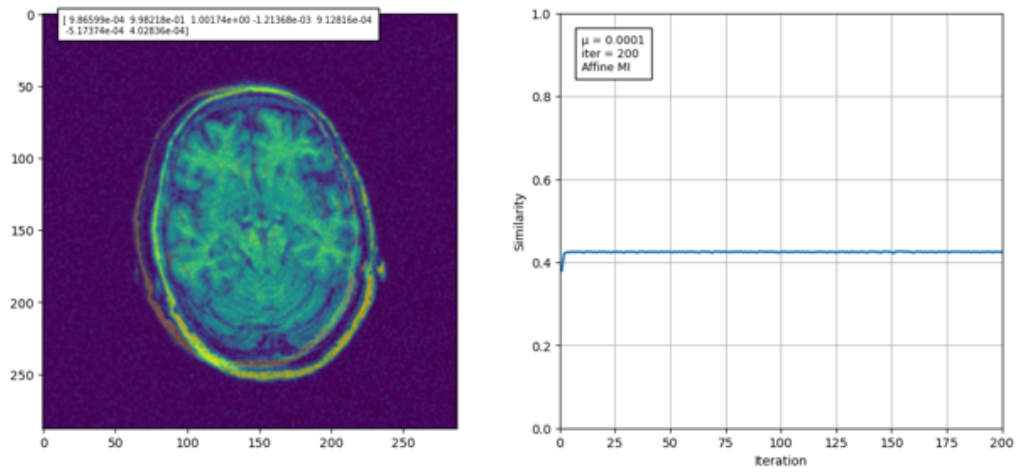


Figure 15: Affine NCC registration after Gaussian noise for 3_1_t1

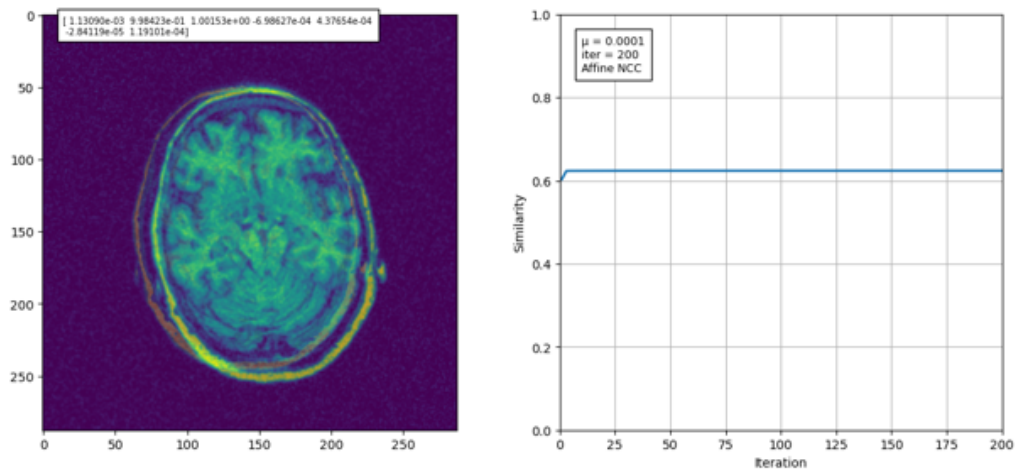


Figure 16: Affine MI registration after Gaussian noise for 3_1_t1

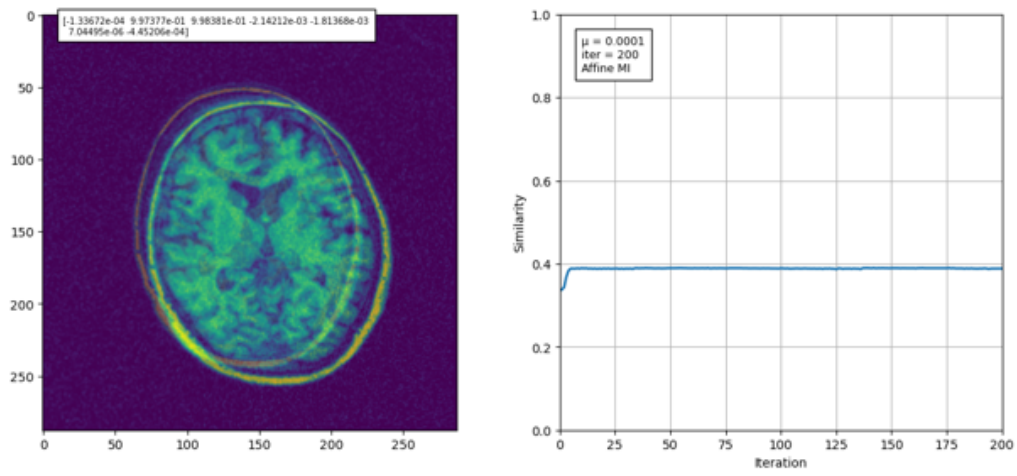


Figure 17: Affine NCC registration after Gaussian noise for 3_2_t1

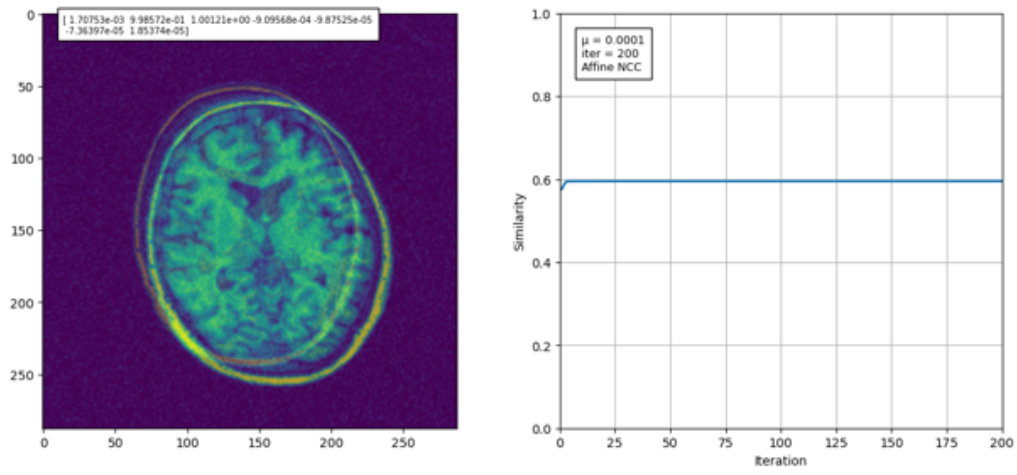


Figure 18: Affine MI registration after Gaussian noise for 3_2_t1